



MINING DEALER BEST PRACTICE SERIES

3500 Engine Series Water Lines Group Parts Salvage

Maintenance and
Repair

Application

Component Life
Management

Component
Renewal (CRC)

MARC
Management

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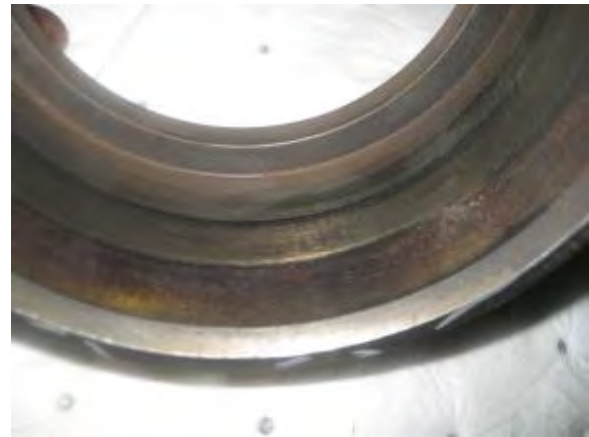
March 2013
1210-4.02.1294

1.0 Introduction

This Best Practice contains information about salvaging water line group parts of 3500 series engines, such as elbows, bonnets and flanges. Salvaging these parts can achieve significant cost savings.

2.0 Best Practice Description

The parts of water line groups, such as flanges, bonnets and elbow tubes, usually have damaged connecting areas. The damaged areas are machined down and fabricated sleeves are installed to enable reusing the parts.



Damage component –Flange 146-7383

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3.0 Implementation Steps

CAUTION: SAFETY PROCEDURES FOR THE SHOP EQUIPMENT INVOLVED MUST BE FOLLOWED.

Implementation steps are defined for salvaging a 146-7383 flange. Similar steps may be to the other parts listed below.

- Make sure the part has been properly cleaned before beginning of the repair
- Inspect the part for cracks. If there is a crack, salvage of the component is not recommended.
- Machine the bore of the flange 146-7383 to 111.2 mm ID; and depth of 48 mm.

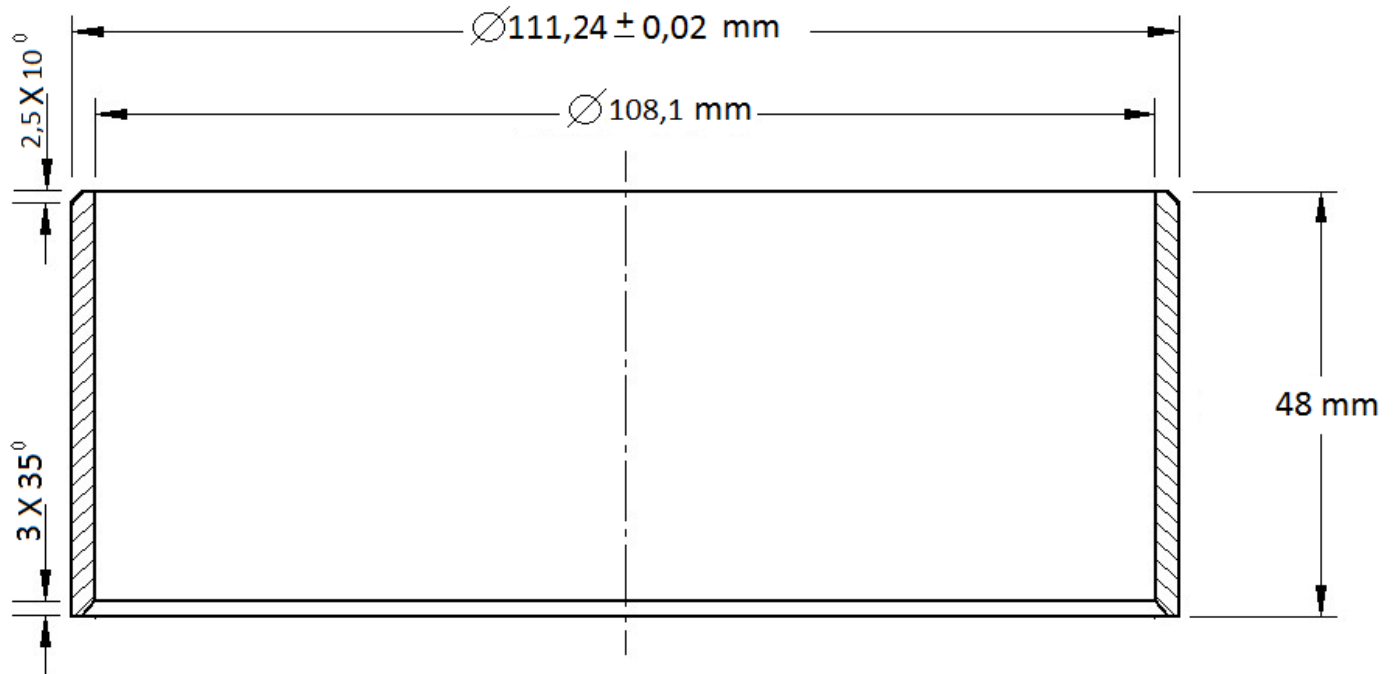


Machining the Flange 146-7383

- Fabricate a sleeve using ASTM A48 -30B (Gray Iron). Detailed dimensions are given below.

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The dimensions are:



A fabricated sleeve

- Cool down the sleeve to -45 degrees C (-50 degrees F) for 2 hours.
- Apply LOCTITE 640 on the machined surface of the part.

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Application of the retaining compound

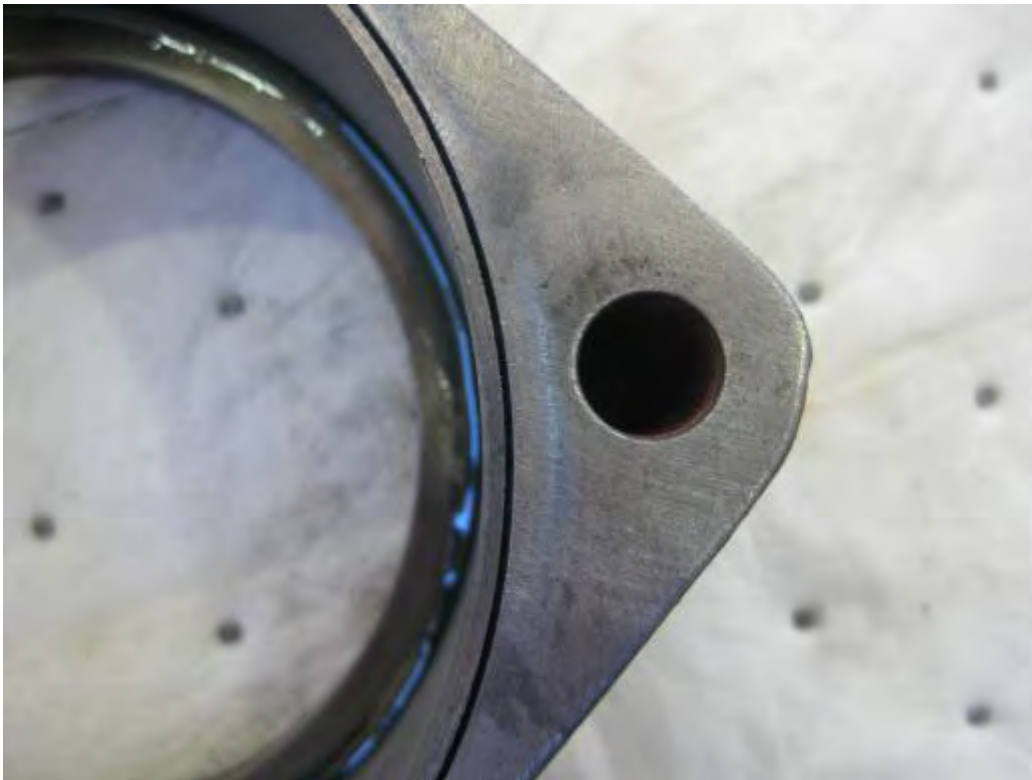
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- Install the cooled sleeve into the machined bore using proper driver adapter.



Installation of the sleeve



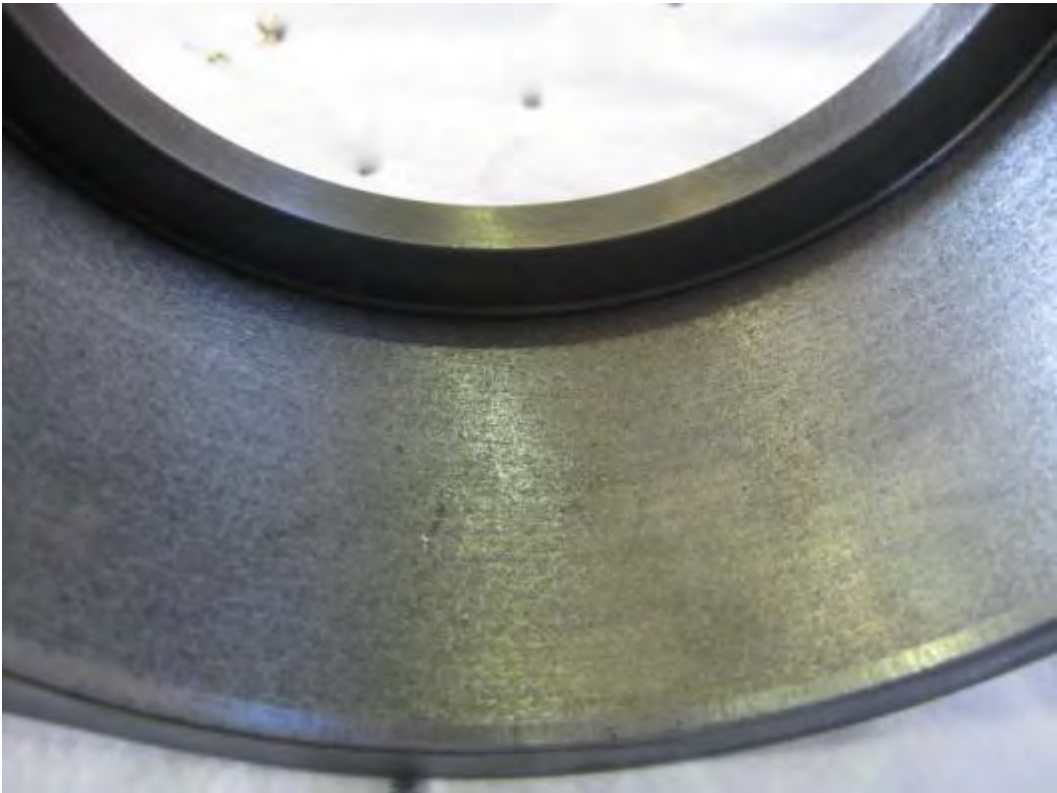
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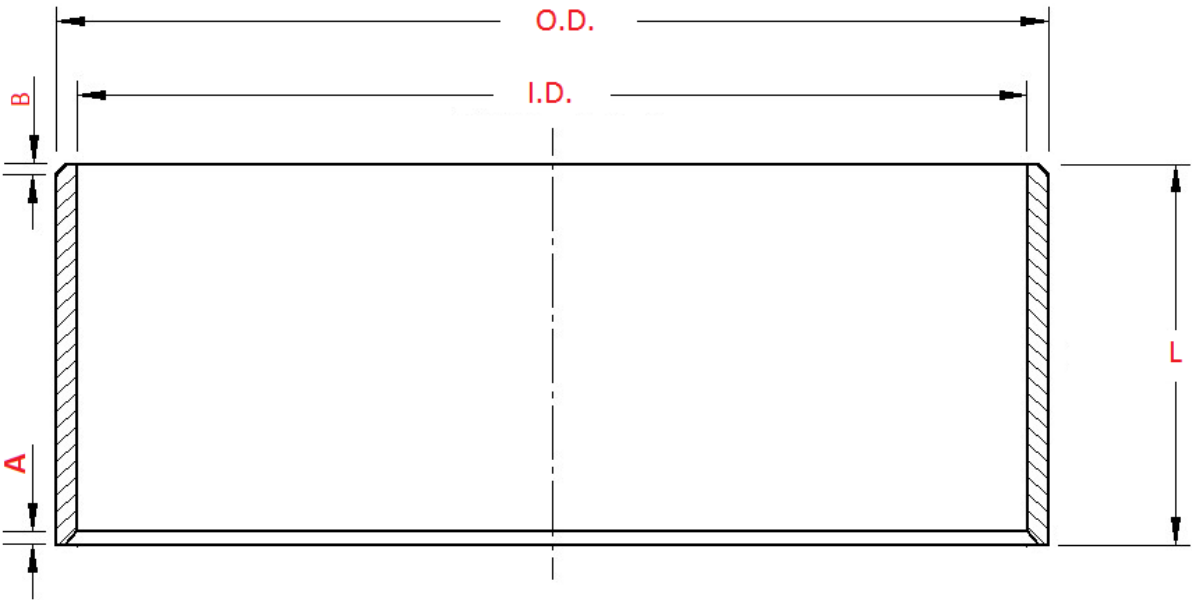
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After salvage

Sleeve dimensions for the parts which can be repaired by applying same method are given on the following table;



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Parts		Dimensions					Sleeve Material	
		Sleeve ID (mm)	Sleeve OD (mm)	Part Counter-bore ID (mm)	Sleeve L (mm)	Sleeve A (mm)		
4W-6077 Elbow		84.45	88.04 ± .02	88	25	3X30 ⁰	2.5X10 ⁰	ASTM A48 - 30B
								
7E-9584 Elbow		97.5	101.04 ± .02	101	30	3X30 ⁰	2.5X10 ⁰	ASTM A48 - 30B
								

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4W-9625 Bonnet	  	1) 84.45 2) 85.90	1) 88.04 ± .02 2) 90.04 ± .02	1) 88 2) 90	25	3X30°	2.5X10°	ASTM A48 - 30B
2W-1055 Elbow	 	65.40	69.04 ± .02	69	30	2.5X30°	2 X10°	ASTM A48 - 30B

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4P-5752 Elbow	 	97.15	101.04 ± .02	101	30	3X30 ⁰	2.5X10 ⁰	ASTM A48 - 30B
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4.0 Benefits

- Cost saving on engine overhauls
- Overhaul turnaround time can be reduced for an engine rebuild, salvaged parts can be inventoried.

5.0 Resources Required

- Machining Center or similar tool to machine part counterbore
- Lathe machine to machine sleeve
- Band saw
- Parts Freezer
- Retaining compound (recommend Loctite 640)
- Sleeve installation driver adapters

6.0 Supporting Attachments / References

EDITOR'S NOTE: While the procedure noted in this Best Practice may work well for some dealers, Caterpillar's preference is to salvage the bore with flame spray and machine to size. The inserts can come loose during service and or leak, however, dealers with no flame spray capabilities will typically install sleeves. See attached SEBF-8148 document for more information.

7.0 Related Best Practices

N/A

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8.0 Acknowledgements

The idea of creating and implementing this practice belongs to:

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